

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

B.Tech Degree S4 (S, FE) / S2 (PT) (S, FE) / S4 (WP) (S) Examination December 2024 (2019 Scheme)

Course Code: EET204**Course Name: ELECTROMAGNETIC THEORY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

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| 1 | Express the point P (4, 90°, 30°) in cartesian coordinates. | 3 |
| 2 | Explain the physical significance of divergence of a vector field. | 3 |
| 3 | What is an equipotential surface? | 3 |
| 4 | Why electric field is conservative? | 3 |
| 5 | What is continuity equation of current? | 3 |
| 6 | Find the magnetic flux density for the vector magnetic potential,
$\vec{A} = e^{-x} \sin y \vec{a}_x + (1 + \cos y) \vec{a}_y$ | 3 |
| 7 | Define propagation constant | 3 |
| 8 | What is Poynting vector? | 3 |
| 9 | What is reflection coefficient of transmission line? | 3 |
| 10 | A loss-less transmission line has characteristics impedance of 70Ω and a phase constant of 3 rad/m at 100 MHz. Calculate the inductance per meter and capacitance per meter of the line. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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| 11 | a) State and explain divergence theorem. | 5 |
| | b) Vector $\vec{A} = \rho \cos \theta \vec{a}_\rho + \rho z^2 \sin \theta \vec{a}_z$ i) Transform $\vec{A}$ into rectangular coordinates and calculate its magnitude at (3, -4, 0). ii) Transform $\vec{A}$ into spherical coordinates and calculate its magnitude at (3, -4, 0). | 9 |
| 12 | a) Evaluate both sides of Stoke's theorem for the field $\vec{G} = 4x\vec{a}_x + 2x^2y\vec{a}_y$ and rectangular path around the region $1 \leq x \leq 2, -1 \leq y \leq 2, z = 0$. Let the positive direction of dS be $\vec{a}_z$ | 10 |
| | b) Define gradient of a scalar function. | 4 |

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Module -2

- 13 a) If $V = x^2y + \frac{2}{x^2+y^2} - 40z$, find $\mathbf{E}$, $\mathbf{D}$, & ρ_v at P (1, 2, 3) 7
- b) State Gauss's law & explain any one application of Gauss's law 7
- 14 a) Derive an expression for electric field intensity at a point due to an electric dipole 10
- b) Find the capacitance of a parallel plate capacitor a) when the plates are of area 1m^2 , distance between the plates 1 mm, voltage gradient is 10^5 V/m and the $\rho_s = 2\text{ }\mu\text{C/m}^2$. b) When the stored energy is 5mJ and the voltage across the plates is 5V. 4

Module -3

- 15 a) Two extensive homogeneous isotropic dielectrics meet on a plane $z = 0$. Region 1 ($z < 0$) contains a dielectric for which $\epsilon_{r1} = 2.5$ while region 2 ($z > 0$) is characterized by $\epsilon_{r2} = 4$. A uniform electric field $\mathbf{E}_1 = -30\bar{a}_x + 50\bar{a}_y + 70\bar{a}_z\text{ V/m}$. Find i) $\mathbf{E}_2$ ii) $\mathbf{D}_2$ iii) Angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the normal. iv) Angles $\mathbf{E}_1$ and $\mathbf{E}_2$ make with the interface. 10
- b) Write the Maxwell's equations for time varying fields in integral form. 4
- 16 a) Let $\mathbf{A} = (y+z)\mathbf{a}_x + 4xz\mathbf{a}_y$ find vector magnetic potential, magnetic flux density, magnetic field intensity & current density at point P (2, -2, 4) 10
- b) Compare conduction current and displacement current 4

Module -4

- 17 a) A lossy dielectric material is characterized by $\mu = 2\mu_0$, $\epsilon = 2\epsilon_0$, and $\sigma = \frac{0.035}{\text{m}}$. If electric field $\bar{E} = 3e^{-\gamma z}\bar{a}_y\text{ V/m}$. At a frequency of 30MHz, determine i) propagation constant ii) Intrinsic impedance iii) Magnetic field of the wave. 10
- b) Explain the significance of intrinsic impedance 4
- 18 a) A uniform plane wave with 10 MHz frequency has average Poynting vector 1W/m^2 . If the medium is perfect dielectric with $\mu_r = 2$ and $\epsilon_r = 3$. Find (i) velocity, (ii) wavelength, (iii) intrinsic impedance 8
- b) What is a lossy dielectric? 6

Module -5

- 19 a) A distortion less transmission line operating at 250MHz has $R = 30\Omega/\text{m}$, $L = 200\text{ nH/m}$ and $C = 80\text{pF/m}$. Determine the i) characteristics impedance ii) Propagation constant iii) Velocity of propagation along the line. 8

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- b) Define standing wave ratio? What is the value of standing wave ratio for i) Short circuited line ii) Open-circuited line iii) a line terminated by its characteristics impedance. 6
- 20 a) Derive an equation for voltage and current at any point of transmission line 10
- b) A lossless transmission line operating at 4.5GHz has $L = 2.4\mu H/m$ and characteristic impedance $Z_0 = 85\Omega$. Calculate phase constant and phase velocity. 4
